

Introduction to Demographic Methods

(1.2: Introduction – additions)

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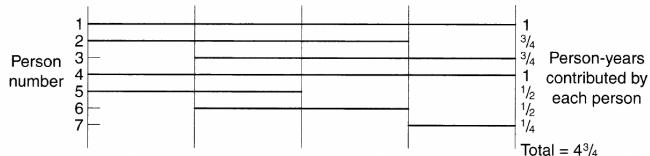
Additional topics, figures, and links for topic 1 (introduction)

- ▶ In demography, it refers to changes in overall population rates (like death, birth, or productivity) driven by shifts in the structure of the population —such as age, gender, or education— rather than changes in the behavior of specific subgroups.
- ▶ These effects, also known as demographic structure impacts or population makeup changes, explain how an aging population increases crude death rates even if medical care improves.
- ▶ Usage Examples in Demography
 - ▶ Demographic Dividend: An increase in the share of the working-age population (a compositional change) can boost economic output per capita.
 - ▶ Mortality Rates: A country with a higher proportion of elderly people may have a higher crude death rate, even if it has better healthcare than a younger country.

Person years contributed from Preston et al., 2000

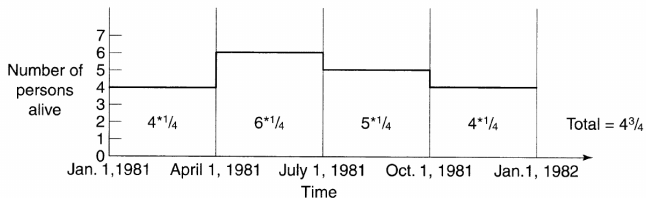


Each horizontal line represents one life line for a specific individual.



a. Life-lines for seven individuals who live in a population at any time between Jan. 1, 1981 and Dec. 31, 1981

Each vertical line represents a quarter in the year.



b. Life-lines converted into numbers of persons alive at each moment

Figure 1.1 Demonstration of the equivalence of the two methods for recording person-years

Importance of using denominators to calculate rates

It is important to use proper “denominators” to calculate rates. This helps put raw counts in a correct perspective.

See one example in this video from national news where the 6 digit precise count of babies born in 2025 is given which is reduced by 3.4% in comparison to 2024.

The first question is, what is the denominator here? Is it based on the number of women in reproductive ages? Did we have a change, increase/decrease, in their population?



See more in: <https://www.youtube.com/watch?v=2l-W7M78Eog&t=724s>

Population pyramid of Mecklenburg-Vorpommern (MV) in 2024

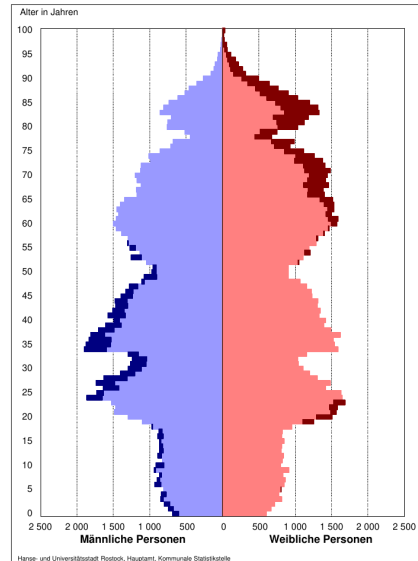


Abb. 2.05 Altersstruktur der Bevölkerung der Hanse- und Universitätsstadt Rostock am 31. Dezember 2024

Each horizontal line represents one age group and the count of men (left, blue) or women (right, red) in that group.

See more on page 41 of:

https://rathaus.rostock.de/de/rathaus/aktuelles_medien/statistisches_jahrbuch_2025_erschienenen/371689

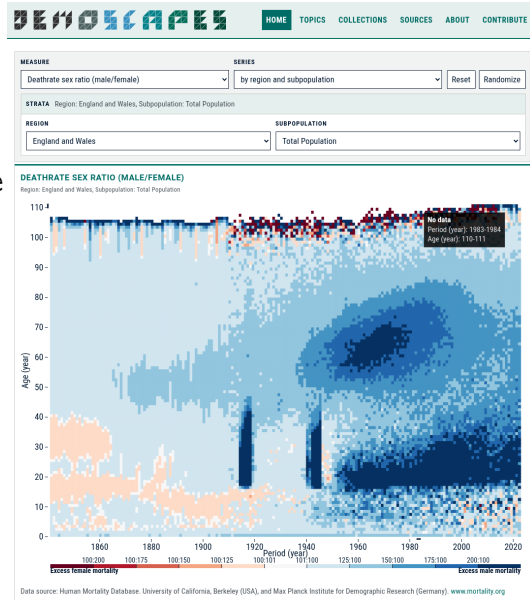


Lexis diagram, and Lexis surface

Visit this website for a collection of Lexis diagram and Lexis surfaces (what was their difference according to Wachter, 2014 book's chapter 2?!):

One example on the right, see more here:

<https://demoscapes.org>



What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the “key reading material”.
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

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References



Preston, S., Heuveline, P., & Guillot, M. (2000). Demography: Measuring and Modeling Population Processes. Wiley.



Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.



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